# Operations Management
# Case: Just Baked™ Inventory Management—Exercise

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**SECTION 2**

**COURSE – Operations Management - II**

## 1. Case Background

Just Baked started in Pam Turkin's kitchen in Livonia, Michigan. She was making s'more cupcakes for her children, word spread, and she began supplying to office events. By early 2013 the company had 13 stores, most of them run by franchisees. All baking still happens at the single Livonia facility, and cupcakes are delivered to every store before opening.

The name is also the operating constraint. Anything unsold at closing time is thrown out. Average selling price is about $2.75. Stores are charged a transfer price of $1.25, but the bakery's real variable cost is only $0.50 (25c material, 25c labour). Discarded cupcakes earn the store a $0.75 credit. Demand moves with season, store location, day of week and time of day, and franchisees are not allowed to place their own orders.

2 packaged cake mixes, vanilla and chocolate, sit behind every recipe. They are ordered weekly from a grocery warehouse with a 3-day lead time. Eggs and butter run on continuous review with a 1-day lead time and cold storage.

## 2. Key Inventory Challenges

- Cupcakes have a 1-day life and no salvage value. Every forecasting error is paid for the same day.
- Demand varies on 4 dimensions, but production is committed centrally before any of that day's demand is seen.
- The $1.25 transfer price makes franchisee economics different from chain economics, so the 2 would stock to different levels.
- The grocery warehouse sometimes cannot supply. The fallback is buying off retail shelves across about 20 stores, which costs roughly $600 each time (15 hours overtime at $20, plus a $300 premium/pallet).
- Eggs have CV = 400/1600 = 0.25 and are critical to production. High variability plus a high required service level means a large buffer inventory.

## 3. Inventory Model for Each Item

The 3 items land in 3 different models, and the deciding factor is what happens to leftover stock, not how perishable the item is.

| Item | Model | Why |
|---|---|---|
| Cupcakes | Single period | Nothing carries into the next day, so the whole decision is one critical ratio, $P \leq Cs / (Ce + Cs)$. |
| Cake mix | P model (periodic review) | Review time is fixed at 1 week and order quantity varies. Shelf life is 18 months and warehouse space is free, so holding is cheap. |
| Eggs | Q model (continuous review) | Stock is checked continuously and an order is triggered at the reorder point. There is a $40 ordering cost and steady usage, so EOQ and ROP apply. |

Worth noting that cupcakes and eggs are both high CV items under XYZ analysis, but they still take different models. Perishability alone does not decide it.

## 4. Analysis

### (a) Cupcakes as a single period problem

The quantity is fixed before demand is known, the selling window is 1 day, and both error directions have a price. Since customers will accept a 2nd choice flavour, Cs is just the lost margin on the sale.

| | Franchisee | Company-owned |
|---|---|---|
| Cs (shortage) = revenue − cost | 2.75 − 1.25 = $1.50 | 2.75 − 0.50 = $2.25 |
| Ce (excess) = cost − salvage | 1.25 − 0.75 = $0.50 | 0.50 − 0 = $0.50 |
| Critical ratio | 1.50 / 2.00 = 75.0% | 2.25 / 2.75 = 81.8% |

81.8% is the number that matters, because it uses the real cost of making a cupcake instead of an internal transfer price.

### (b) Service level for cake mix

Cake mix is periodic review, so the risk window is the exposure period: 7 day review + 3 day lead time = 10 days. Physical holding cost for cake mix is treated as zero at the margin because Livonia has spare space. That stops being true as the chain grows. Cake mix working covers vanilla only, since that is the only demand distribution given. Lead times are treated as fixed. Variable lead times would need more safety stock.

| Parameter | Value |
|---|---|
| Cs (stockout per pallet) = $300 overtime + $300 premium | $600 |
| Annual holding cost, H = i × C = 0.12 × 1000 | $120 /pallet/year |
| Ce (1 review period) = 120 / 52 | $2.31 /pallet/week |
| Critical ratio = 600 / (600 + 2.31) | 0.9962 |

The two costs are nowhere near each other, which is why the service level lands so close to 100%. Reading down the cumulative column for vanilla, 0.99 at 4 pallets is still short of 0.9962, and 0.997 at 5 pallets clears it.

| Pallets | Cumulative probability | Clears 0.9962 ? |
|---|---|---|
| 2 | 0.40 | No |
| 3 | 0.95 | No |
| 4 | 0.99 | No |
| 5 | 0.997 | Yes, so OUTL = 5 |

Target stock level is 5 pallets. With 2 pallets on hand at the review, Q* = 5 − 2 = 3 pallets.

### (c) Why eggs are not a single period item

Leftover eggs get used the next morning. That is the whole difference. A cupcake left at closing is worth nothing, an egg left in cold storage is worth exactly what it was worth yesterday, so the period boundary does not destroy value and the critical ratio does not apply.

Perishability still shows up, just as a holding cost of $0.02/egg/day rather than a write off. Combined with the $40 ordering cost and continuous usage, that gives an ordering cost versus holding cost trade-off, which is EOQ.

| | Working | Result |
|---|---|---|
| EOQ, Q* = √(2DS/H) | √(2 × 1666.7 × 40 / 0.02) | 2,582 eggs |
| Safety stock = zσ, z = 3.00 | 3.00 × 400 | 1,200 eggs |
| ROP = đLT + safety stock | (1600 × 1) + 1200 | 2,800 eggs |
| ROP in days of supply | 2800 / 1600 | 1.75 days |
| Average on hand = Q/2 + SS | 1291 + 1200 | 2,491 eggs, or 1.56 days |

Since the exposure period is one day, σ over the exposure period is just the daily σ, so average days of supply reduces to Q*/2μ + z × CV. At CV = 0.25 that is 0.81 + 0.75 = 1.56 days at three sigma, and 0.81 + 0.50 = 1.31 days at two sigma. Each sigma of cover costs one CV worth of days.

One inconsistency in the case: 6,00,000 eggs over 360 days gives 1,666.7 a day, but the case also says daily demand averages 1,600. EOQ above uses the annual figure, ROP uses the daily one, following the hints. Using 1,600 throughout gives EOQ 2,530 and average stock 2,465, which changes nothing important.

### (d) Why franchisees do not place their own orders

A franchisee optimising for itself stocks to 75%. The chain optimum is 81.8%. Left alone, franchisees would order too few.

The reason is that $1.25 is a real cost to a franchisee but only an internal transfer for Just Baked. The only money that actually leaves the system is the $0.50 it costs to bake. So every cupcake between the 75th and the 81.8th percentile of demand loses the franchisee money and makes the chain money. Central ordering captures those units. It also pools demand across 13 stores instead of forecasting each one separately, which cuts the relative variability the bakery has to cover.

## 5. Managerial Recommendation

- Keep ordering central and stock to 81.8%. If Just Baked ever wants franchisees to order locally, fix the contract first. Holding the transfer price at $1.25, the discard credit would need to go from $0.75 to about $0.92 for the franchisee's own optimum to land at 81.8%. Cutting the transfer price does the same job from the other side.

- Treat the cake mix problem as a supply issue, not just a stocking one. A 99.6% service level is the right answer given the costs, since a stockout runs about 260 times the weekly cost of holding a spare pallet, and OUTL of 5 follows. But a second supplier or a firm commitment from the warehouse removes the $600 event instead of insuring against it.

- For eggs, work on demand variability before touching the service level. Safety stock is 1,200 of the 2,491 eggs held on average, and that is entirely a CV = 0.25 problem. Dropping from three sigma to two saves 400 eggs, about $8 a day, while raising stockout risk from 0.13% to 2.3% on an input without which nothing gets baked. Better day of week forecasting is the cheaper fix.

- As the chain grows past 13 stores, the 3 items will need 3 separate routines. Right now they run on roughly the same weekly rhythm.

- Unsold cupcakes are worth another look. At 81.8% there is leftover stock most days, and the cost of it scales with store count. Late day discounting and inter-store transfers during the day are both worth testing. None of it should be allowed to weaken the freshness claim.

## PDF Answers –

**Q1. Best service level at each store**

| | Franchisee | Company-owned |
|---|---|---|
| a) Cs = price − cost | 2.75 − 1.25 = $1.50 | 2.75 − 0.50 = $2.25 |
| b) Ce = cost − credit | 1.25 − 0.75 = $0.50 | 0.50 − 0 = $0.50 |
| c) and d) SL* = Cs/(Cs+Ce) | 1.50/2.00 = 75.0% | 2.25/2.75 = 81.8% |

e) Franchisees would stock to 75%, the chain optimum is 81.8%, so decentralised ordering leaves the units between those two percentiles unsold. Those units are profitable for Just Baked because the real cost is $0.50, not $1.25.

**Q2. Exposure period, cake mix**

a) Review period = 7 days, since planning is weekly
b) Lead time = 3 days from order to delivery
c) Exposure period = 7 + 3 = 10 days

**Q3. Stockout and holding costs, cake mix**

a) Cs = extra labour + extra material = (15 × 20) + 300 = $600 per pallet
b) H = i × C = 0.12 × 1000 = $120 per pallet per year
c) Ce through one review period = 120 / 52 = $2.31 per pallet per week
d) SL* = 600 / (600 + 2.31) = 0.9962, or 99.62%

**Q4. Target stock and ordering, vanilla cake mix**

a) Find the first cumulative probability greater than 0.9962. At 4 pallets it is 0.99, which is short. At 5 pallets it is 0.997, which clears it.

| Pallets | Probability | Cumulative | ≥ SL*? |
|---|---|---|---|
| Fewer than 2 | 0.10 | 0.10 | No |
| 2 | 0.30 | 0.40 | No |
| 3 | 0.55 | 0.95 | No |
| 4 | 0.04 | 0.99 | No |
| 5 | 0.007 | 0.997 | Yes, OUTL = 5 |
| 6 | 0.003 | 1.00 | Yes |

Target stock level (OUTL) = 5 pallets
b) Q* = OUTL − Inet = 5 − 2 = 3 pallets

**Q5. Ordering eggs**

Converting to daily values as the hint directs:
d = 600,000 / 360 = 1,666.7 eggs per day
H = $0.02 per egg per day
S = $40

| | Working | Answer |
|---|---|---|
| a) EOQ, Q* = √(2DS/H) | √(2 × 1666.7 × 40 / 0.02) | 2,582 eggs |
| b) Safety stock = zσ, z = 3.00 | 3.00 × 400 | 1,200 eggs |
| c) ROP = đLT + SS | (1600 × 1) + 1200 | 2,800 eggs |
| d) X = ROP / daily demand | 2800 / 1600 | X = 1.75 days |
| e) Iavg = Q/2 + SS | 1291 + 1200 | 2,491 eggs = 1.56 days |

f) SS = zSL × σEP, and the exposure period for eggs is one day
so σEP = σ and SS/μ = zSL × CV. Average on-hand days of supply is therefore Q*/2μ + zSL × CV. With CV = 400/1600 = 0.25 and Q*/2μ = 2582/3200 = 0.81, dropping from 3-sigma to 2-sigma gives 0.81 + 2(0.25) = 1.31 days, against 1.56 days at 3-sigma.
The saving is one CV worth of days, or 400 eggs.